KUSANAPELLI SAI KRUPAL

AI-Assisted Builder & Operator | Backend & Systems | B.Tech CSE (2025)
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SUMMARY

2025 B.Tech Computer Science graduate (GNIT Hyderabad). Founder and AI-assisted builder/operator of a live multi-tier student-services platform (Next.js frontend, Node.js API layer, PostgreSQL/Supabase, Vercel) used by 150+ students across 2 colleges, run as a registered business — I own the integration, deployment, debugging and daily operations: RESTful API routes for auth, payments and notifications, relational database schema design and migrations, webhook-verified payment flows, Git-based deployments, and a production voice-calling backend (WebRTC media server on an Oracle Cloud Linux VM, speech-to-text and Telugu text-to-speech wired end to end). Computer-science fundamentals — Python programming (daily hands-on practice), object-oriented design, data structures and algorithms; SQL and REST APIs at working level from production.

TECHNICAL SKILLS

- **Languages:** Python — daily hands-on practice; JavaScript/TypeScript — working knowledge, used daily in production (AI-assisted coding)
- **Backend:** Node.js (Next.js API routes — 152 RESTful API routes designed and operated for auth, payments, notifications); webhook-based payment verification (Razorpay live; Cashfree, PhonePe integrated)
- **Auth & security:** custom session authentication — hashed tokens, sliding expiry, role-based allow-list authorization in application code; HMAC-signed webhooks verified before processing; API rate limiting
- **Data:** PostgreSQL via Supabase — relational schema design, SQL queries, 100+ migrations operated in production
- **Voice & AI:** WebRTC media server on an Oracle Cloud Linux VM — speech-to-text (Groq Whisper) → answer engine → Telugu text-to-speech (Piper); LLM provider fallback chain (Gemini → Groq → OpenRouter → Cloudflare) answering support questions grounded in real student records
- **Tools & ops:** Git/GitHub (version control, deployments), Vercel (hosting, cron), Linux daily operations (SSH, logs, systemd services), Android app via Capacitor; live production debugging and incident triage; support operations — 1,300+ real user messages; web push notifications
- **Cloud (AWS):** root account created and console access verified (Free Tier)

EXPERIENCE

Founder & Operator — Student Platform

Live in production | 2026 – Present

- Full-stack multi-tier web application (Next.js frontend, Node.js API routes, PostgreSQL database, Vercel hosting) used by **150+ students** across 2 colleges: registration, document submission, fee payment, no-due certificates, High Court case tracking with hearing alerts.
- **PWA & case tracking:** the platform runs as an installable PWA; live High Court case status auto-fetched by case number, with hearing alerts for 150+ students.
- **Payments:** Razorpay live — order creation, checkout, HMAC-verified webhook → idempotent approval; the app charges one total fee of ■2,200 and accepts automated gateway verification only (manual screenshot approvals refused by design); Cashfree and PhonePe integrations wired behind gateway switching.
- **Voice backend:** designed and deployed a live in-app calling backend — WebRTC media server on an Oracle Cloud Linux VM with an HMAC-signed negotiation endpoint; browser mic audio → speech-to-text → answer engine → Telugu text-to-speech, verified end to end; diagnosed and fixed a systemd crash-loop by tracing service logs to a missing environment file.
- **Support & AI:** WhatsApp/Telegram support bot with conversation context; AI answers grounded in each student's own records with a provider fallback chain and human escalation instead of invented answers; 1,300+ real user messages handled.
- **RAG assistant:** built a retrieval-augmented support assistant (LangChain + ChromaDB + local sentence-transformer embeddings) answering platform help questions from a hand-written corpus, with a similarity threshold that rejects off-topic questions instead of inventing answers; retrieval verified 6/6 tests.
- **Operations:** Git for all version control; deployments via Vercel, verified on the live site with real data before being called done; production incident triage for live users (e.g. restored an admin students list broken by an ambiguous database relationship, verified across all admin roles).

EDUCATION

B.Tech, Computer Science and Engineering | 2025

Guru Nanak Institute of Technology (GNIT), Hyderabad, Telangana